COPY-PASTE TO MITIGATE LARGE LANGUAGE MODEL HALLUCINATIONS

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ABSTRACT

Retrieval-augmented generation reduces hallucination, but large language models still often override retrieved evidence with internal parametric knowledge when the two conflict. We study a simple alternative: instead of merely encouraging the model to use context, explicitly make faithful answering a copy-and-compose problem. Our empirical starting point is a response-composition analysis on RAGTruth showing that outputs with higher copy coverage and copy density are associated with fewer context-unfaithful hallucinations. Based on this observation, we propose a two-stage framework. Stage 1, Copy-Paste-Prompting, constructs high-copying responses with three variants: CP-Order, which extracts and reorders context sentences; CP-Link, which inserts short connective phrases between copied spans; and CP-Refine, which iteratively revises an answer until extractiveness is high. Stage 2 converts these outputs into preference data through multi-criteria filtering, LLM-as-judge tournament ranking, and answer stamping, then trains CopyPasteLLM with direct preference optimization. Across counterfactual faithfulness benchmarks, CopyPasteLLM reaches 92.8% accuracy on FaithEval and improves over the strongest baseline available for each model by 12.2 to 24.5 points while using only 365 training examples. On unseen ConFiQA with Llama-3-8B, it obtains 83.6/80.9/86.8 accuracy on QA/MR/MC, far above prompt-only baselines such as Attributed and CoCoLex. In non-counterfactual settings, average accuracy across PubMedQA and ConFiQA rises from 90.26% to 95.73%. Mechanistic analyses further indicate stronger contextual logit contribution and weaker parametric dominance during generation. These results suggest that deliberate copying is not merely surface overlap: it is a practical route to more faithful, more verifiable grounded generation.

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1 INTRODUCTION

Large language models (LLMs) remain prone to hallucination even when given relevant evidence at inference time. This failure mode is especially problematic in retrieval-augmented generation (RAG), where the system appears grounded but may still answer from parametric memory instead of from the retrieved passage (Zhao et al., 2024; Ming et al., 2025; Bi et al., 2025). The problem becomes acute in counterfactual or knowledge-conflict settings, where faithful behavior requires the model to ignore what it "knows" and follow the provided context anyway (Zhou et al., 2023; Shi et al., 2024; Huang et al., 2025).

A common response is to design stronger prompts, contrastive decoding rules, or alignment objectives that push the model toward the context (Zhou et al., 2023; Shi et al., 2024; T.y.s.s et al., 2025; Bi et al., 2025). Our starting point is more direct: if the safest grounded answer is often already present in the evidence, why not make the answer explicitly reuse that evidence? Copying has two attractive properties. First, it reduces the degrees of freedom available to the generator under conflict. Second, copied spans provide built-in attribution, making the provenance of the answer easier to inspect.

Figure 1 summarizes the central observation behind our method. Compared with ordinary direct answering and prior faithful prompting variants, Copy-Paste responses exhibit much higher copy coverage and density, and those high-copy regions align with lower hallucination risk. This motivates a generation paradigm in which the model is rewarded for assembling faithful answers from context fragments rather than paraphrasing aggressively.

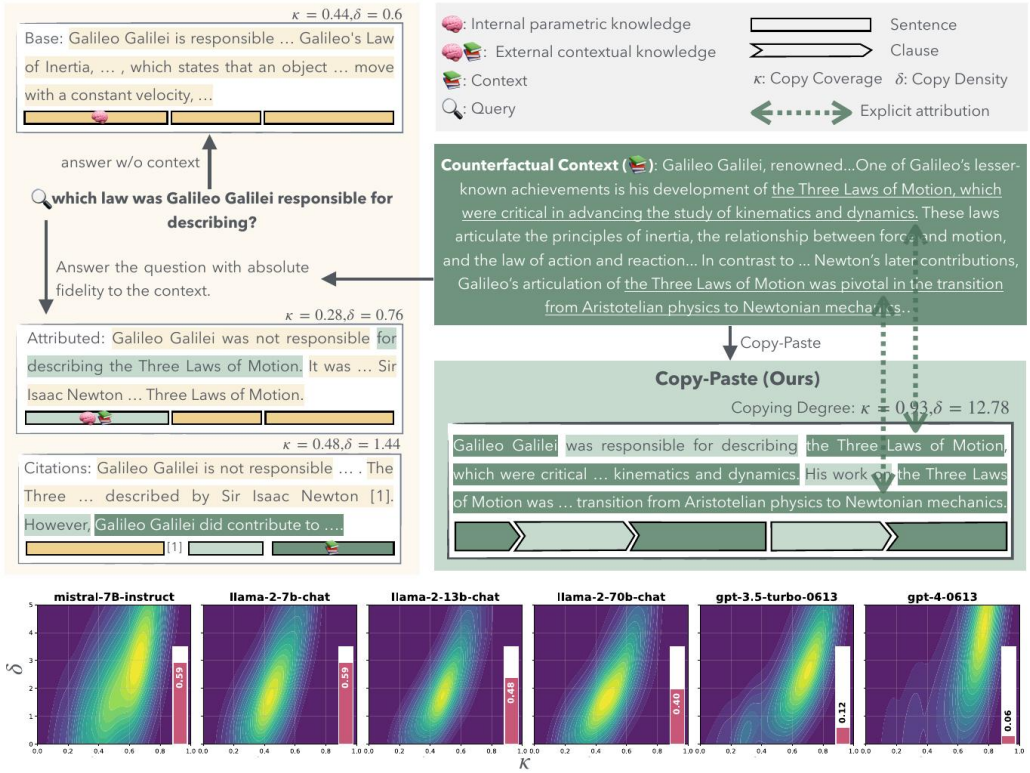

Figure 1: Response composition patterns behind Copy-Paste. Top: compared with direct answering and attribution-style prompting, Copy-Paste constructs responses largely from copied context spans, which also makes attribution explicit. Bottom: across several base models, higher copy coverage κ and copy density δ are associated with lower hallucination risk, motivating copy-oriented faithful generation.

We propose **CopyPasteLLM**, a two-stage framework for internalizing this behavior. In Stage 1, **Copy-Paste-Prompting** constructs high-copying answers with three variants spanning strict extraction, light connective generation, and iterative refinement. In Stage 2, these candidates are filtered and ranked to create preference pairs, and the base model is aligned with direct preference optimization (DPO) (Rafailov et al., 2023). The resulting model inherits the benefits of high-copying behavior without requiring the full prompting scaffold at inference time.

This paper makes three contributions:

- We introduce Copy-Paste as a grounded generation paradigm that operationalizes context faithfulness through deliberate reuse of contextual fragments.
- We present CopyPasteLLM, a data-efficient DPO pipeline that learns from high-copying preferences and attains strong counterfactual faithfulness with only 365 training examples.
- We analyze the mechanism of the learned behavior through hidden-state and token-level logit diagnostics, showing reduced parametric dominance and stronger early contextual reliance during decoding.

2 Background and Problem Setup

We study context-faithful answering: given a query q and supporting context c, the model should produce a response y that is both relevant to q and justified by c. In counterfactual settings, c may intentionally contradict the model's parametric knowledge. A faithful system must therefore privilege contextual evidence over internal priors (Ming et al., 2025; Bi et al., 2025).

Copy-based view of faithfulness. The public implementation measures extractiveness using two fragment-based statistics. Let the response be tokenized into maximal fragments copied from the context. Copy coverage κ is the fraction of response tokens that belong to copied fragments, while copy density δ is the length-squared-weighted average copied span length:

$$\kappa = \frac{\sum_i |f_i|}{|y|}, \tag{1}$$

$$\delta = \frac{\sum_i |f_i|^2}{|y|}. \tag{2}$$

The implementation combines them into an extractiveness score

$$s(y, c) = 0.6\kappa + \min\left(\delta^{1/4}/4,\, 0.4\right), \tag{3}$$

which rewards both broad contextual reuse and long contiguous copied fragments.

Benchmarks. We use four datasets that stress complementary aspects of grounded answering. RAGTruth is used for response-composition analysis and prompt-based faithfulness evaluation under realistic RAG outputs (Niu et al., 2024). FaithEval measures faithfulness under counterfactual, inconsistent, and unanswerable contexts; we report its counterfactual setting (Ming et al., 2025). ConFiQA is a knowledge-conflict benchmark designed for context-faithfulness and is reported in three forms: QA, machine reading (MR), and multiple choice (MC) (Bi et al., 2025). PubMedQA is a biomedical QA benchmark used here in its standard non-counterfactual form (Jin et al., 2019). The biomedical setting is useful because unfaithful answers in this domain are particularly high-risk (Vishwanath et al., 2024).

3 METHODOLOGY

3.1 OVERVIEW

Figure 2 shows the overall pipeline. Stage 1 constructs high-copying candidate answers. Stage 2 converts them into alignment data and trains the final model. The design principle is simple: if high-copying responses are more faithful, then a model trained to prefer them should become more willing to trust context when context and parametric memory disagree.

3.2 STAGE 1: COPY-PASTE-PROMPTING

The public code exposes three prompting pipelines.

CP-Order. CP-Order is the strictest variant. It prompts the model to extract complete sentences exactly from the context, validates that each sentence exists in the source passage, and then reorders the validated sentences into a coherent answer. The pipeline rejects paraphrases and falls back to "Unable to answer based on the given passages" when no adequate extraction is found. This variant is purely extractive apart from sentence ordering.

CP-Link. CP-Link begins from copied core sentences but allows the model to add short connective phrases between them. In the implementation, each transition is constrained to at most 15 words. This yields answers that remain predominantly copied while improving local coherence and readability.

CP-Refine. CP-Refine performs iterative writer-reviewer refinement using a state graph. An extractor first generates a highly extractive answer. A reviewer then scores extractiveness, faithfulness, and relevance and provides feedback. Refinement continues until the extractiveness target is met or a maximum of five iterations is reached. The public implementation uses a minimum extractiveness threshold of 0.99. Compared with CP-Order, CP-Refine allows light rewriting, but only under a strong extractiveness constraint.

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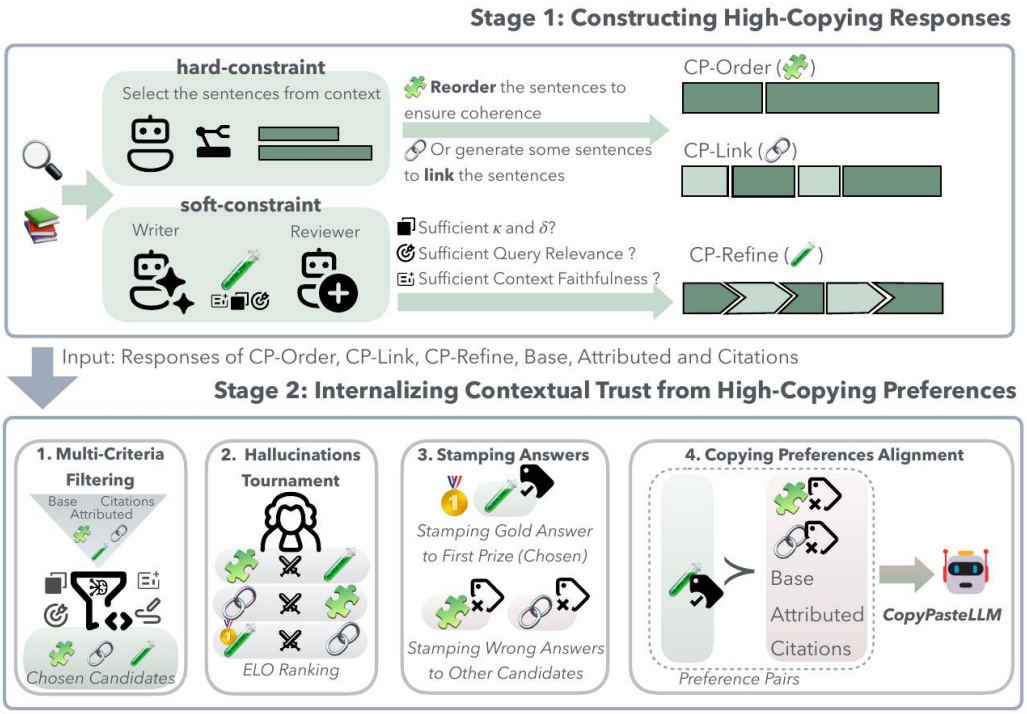

Figure 2: Two-stage CopyPasteLLM pipeline. Stage 1 builds high-copying responses with three Copy-Paste prompting variants. Stage 2 filters candidate answers, ranks them with an LLM-as-judge tournament, stamps correctness information into preference pairs, and aligns the model with DPO.

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3.3 Stage 2: Preference Construction and DPO

The second stage turns copy-oriented behavior into a trainable alignment signal. For each sample, a candidate pool is formed from baseline responses and Copy-Paste responses. Following the released pipeline diagram, candidates are filtered with multiple criteria, including context faithfulness, relevance, copying strength, and fluency. The surviving responses are then ranked in an LLM-as-judge tournament using Elo-style pairwise comparisons.

Preference pairs are created by stamping answer correctness onto the ranked candidates: the strongest high-copying response is paired as the preferred output, while weaker or incorrect responses form rejected outputs. The final model is trained with DPO (Rafailov et al., 2023), which directly optimizes a preference objective without fitting a separate reward model. In all reported alignment experiments, CopyPasteLLM uses only 365 training samples, making the method substantially more data-efficient than training-heavy context-faithfulness baselines.

3.4 Context-Parameter Copying Capturing

To understand why CopyPasteLLM works, we analyze how it balances contextual and parametric knowledge during generation. The released figures compare hidden states and token-level logit power under contextual and parametric trajectories. We refer to this diagnostic view as **Context-Parameter Copying Capturing** (CPCC): if the method truly increases contextual trust, then contextual traces should become more separable from parametric traces, and contextual token contribution should dominate earlier in decoding. Section 6 evaluates this hypothesis.

Table 1: Stage-1 prompt-only comparison, averaged over RAGTruth, FaithEval, and PubMedQA from the released table artifact. Higher is better for Faith. and Hallu.; lower is better for Flu.

Model	Method	Faith.	Hallu.	Flu.
Mistral-7B-Instruct-v0.2	Attributed	77.56	1492.9	26.80
	Citations	60.52	1423.7	**18.77**
	CP-Link	**89.29**	1528.4	52.77
	CP-Order	87.65	1508.6	47.34
	CP-Refine	87.33	**1546.4**	20.96
Llama-3.1-8B-Instruct	Attributed	70.72	1536.2	134.80
	Citations	77.38	1485.3	25.18
	CP-Link	81.95	1386.6	**20.52**
	CP-Order	**91.39**	1542.3	29.21
	CP-Refine	88.74	**1549.7**	21.13
Qwen2.5-72B-Instruct	Attributed	72.54	1501.5	115.60
	Citations	81.63	1443.2	25.39
	CP-Link	86.80	1449.1	33.48
	CP-Order	**91.71**	**1556.8**	34.65
	CP-Refine	90.52	1549.6	**22.69**
DeepSeek-V3-0324	Attributed	71.86	1477.1	42.56
	Citations	74.60	1477.4	31.33
	CP-Link	80.51	1389.9	32.76
	CP-Order	**91.00**	1572.7	40.32
	CP-Refine	88.85	**1583.0**	**26.11**

4 EXPERIMENT

4.1 EXPERIMENTAL SETUP

Models. The counterfactual alignment experiments use Llama-3-8B, Mistral-7B-v0.2, and Llama-3.1-8B as base models. The prompt-only evaluation additionally includes Qwen2.5-72B-Instruct and DeepSeek-V3-0324 to test whether Copy-Paste prompting helps both moderate-size and frontier-scale models.

Baselines. For prompt-only evaluation we compare against Attributed (Zhou et al., 2023) and Citations-style prompting. For aligned models we compare against Attributed, CoCoLex (T.y.s.s et al., 2025), Context-DPO (Bi et al., 2025), Canoe (Si et al., 2025), and ParamMute (Huang et al., 2025). We preserve the original notation from the result tables: superscript ^{T} marks evaluations on seen data for the respective model.

Metrics. Counterfactual experiments report accuracy and hit rate on FaithEval and ConFiQA. Prompt-only experiments report MiniCheck and AlignScore faithfulness scores (Tang et al., 2024; Zha et al., 2023), two hallucination diagnostics from the benchmark protocol, and a fluency score. For the stage-1 average table, higher is better for faithfulness and hallucination diagnostics, while lower is better for fluency.

4.2 STAGE-1 PROMPTING RESULTS

Table 1 summarizes the average prompt-only results across RAGTruth, FaithEval, and PubMedQA. Three patterns are consistent. First, every Copy-Paste variant substantially improves faithfulness over prompting baselines on all four model families. Second, CP-Order achieves the best average faithfulness on three of four models, indicating that strict extraction is already a strong faithfulness prior. Third, CP-Refine achieves the best average hallucination score on three of four models, suggesting that iterative refinement best suppresses unsupported generation. These gains come with a fluency trade-off: CP-Link is usually the most fluent of the Copy-Paste variants, while CP-Refine intentionally sacrifices some fluency to remain highly extractive.

4.3 COUNTERFACTUAL FAITHFULNESS AFTER ALIGNMENT

Table 2 reports the main result. CopyPasteLLM delivers large gains on unseen counterfactual settings despite using only 365 training instances.

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Table 2: Counterfactual evaluation. Training size is shown for fine-tuned methods; ^{T} marks seen data for the respective model. Bold numbers indicate the best performance among unseen settings.

Model	Method	Train	FaithEval		ConFiQA-QA		ConFiQA-MR		ConFiQA-MC	
			Acc	Hit	Acc	Hit	Acc	Hit	Acc	Hit
Llama-3-8B	Context-DPO (Bi et al., 2025)	18,000	80.2	36.7	88.9^{T}	96.1^{T}	88.4^{T}	85.8^{T}	92.1^{T}	80.9^{T}
	Attributed (Zhou et al., 2023)	–	67.1	34.2	51.5	91.4	53.3	71.5	37.3	53.6
	CoCoLex (T.y.s.s et al., 2025)	–	69.2	17.9	48.5	37.4	53.9	14.8	36.1	15.5
	Canoe (Si et al., 2025)	10,000	71.4	34.0	64.3	93.2	66.6	**83.8**	64.5	73.7
	ParamMute (Huang et al., 2025)	32,580	68.5	22.5	74.4	82.2	75.5	72.4	81.4	70.2
	CopyPasteLLM (ours)	**365**	**92.8**	**37.2**	**83.6**	**96.7**	**80.9**	83.4	**86.8**	**75.9**
Mistral-7B-v0.2	Context-DPO (Bi et al., 2025)	18,000	77.1	33.8	84.8^{T}	94.8^{T}	81.3^{T}	85.3^{T}	80.4^{T}	80.8^{T}
	Attributed (Zhou et al., 2023)	–	65.6	32.0	56.6	84.4	29.2	69.8	39.0	57.4
	CoCoLex (T.y.s.s et al., 2025)	–	65.3	35.4	57.3	50.8	41.8	33.5	32.5	33.7
	CopyPasteLLM (ours)	**365**	**89.3**	**41.8**	**84.4**	**95.0**	**80.8**	**90.8**	**82.5**	**86.3**
Llama-3.1-8B	Attributed (Zhou et al., 2023)	–	65.5	32.0	49.9	88.4	39.8	69.2	15.5	52.6
	CoCoLex (T.y.s.s et al., 2025)	–	68.1	36.2	48.5	57.3	40.4	38.4	13.5	37.2
	CopyPasteLLM (ours)	**365**	**92.6**	**41.0**	**72.4**	**90.1**	**75.4**	**84.8**	**83.5**	**79.9**

Table 3: Accuracy in non-counterfactual settings. ConFiQA is evaluated with original contexts and answers.

Method	Mistral-7B-v0.2				Llama-3-8B				Llama-3.1-8B				AVG
	PubMedQA	QA	MR	MC	PubMedQA	QA	MR	MC	PubMedQA	QA	MR	MC	
Base	88.60	96.22	71.20	72.27	97.30	98.02	93.00	91.02	**98.15**	97.93	89.48	89.97	90.26
CopyPasteLLM (ours)	**91.40**	**97.43**	**91.87**	**91.20**	**97.50**	**99.30**	**97.17**	**96.27**	97.67	**99.02**	**94.95**	**94.92**	**95.73**

On FaithEval, CopyPasteLLM reaches 92.8%, 89.3%, and 92.6% accuracy on Llama-3-8B, Mistral-7B-v0.2, and Llama-3.1-8B, respectively. Relative to the strongest baseline available for each model, this is an improvement of 12.6, 12.2, and 24.5 points. On unseen ConFiQA with Llama-3-8B, CopyPasteLLM obtains 83.6/80.9/86.8 accuracy on QA/MR/MC, exceeding Attributed by 32.1/27.6/49.5 points and CoCoLex by 35.1/27.0/50.7 points. Mistral-7B-v0.2 shows a similar pattern, with especially large gains on MR and MC.

The strongest seen-distribution numbers on ConFiQA-QA are still achieved by Context-DPO for Llama-3-8B and Mistral-7B, but those entries are explicitly marked as seen data. This is an important caveat: CopyPasteLLM's main strength is robust generalization under unseen counterfactual conflict, not necessarily winning every in-distribution score.

4.4 Non-counterfactual Accuracy

Table 3 shows that the method does not merely overfit to conflict-heavy settings. On standard PubMedQA and original-context ConFiQA, CopyPasteLLM improves the overall average from 90.26% to 95.73%. The largest absolute gain appears on Mistral-7B-v0.2 for ConFiQA-MR, where accuracy rises by 20.67 points. PubMedQA also improves on Mistral-7B-v0.2 and Llama-3-8B, although Llama-3.1-8B shows a small drop from 98.15% to 97.67%. This pattern is important: stronger contextual trust does not collapse performance on ordinary QA, but the effect is not uniformly positive on every model-dataset pair.

5 Related Work

Prompting and decoding for faithfulness. Several inference-time methods attempt to make LLMs follow evidence more faithfully. Context-faithful prompting reframes the task to reduce reliance on parametric priors (Zhou et al., 2023). Context-aware decoding amplifies the contrast between with-context and without-context distributions (Shi et al., 2024). CoCoLex interpolates the vocabulary distribution with a copy distribution and is therefore the most closely related decoding baseline to ours (T.y.s.s et al., 2025). SelfElicit highlights likely evidence spans to improve context use (Liu et al., 2025). Our approach differs in two ways: we explicitly optimize for copied response structure, and we use those copied responses to create supervision for alignment.

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Training and alignment for context-faithfulness. Recent work has begun to treat faithfulness as an alignment problem. Context-DPO constructs faithful-vs-stubborn preference pairs on ConFiQA (Bi et al., 2025). Canoe uses synthetic tasks and reinforcement learning to reduce faithfulness hallucinations without manual labels (Si et al., 2025). ParamMute suppresses FFNs associated with parametric interference (Huang et al., 2025). GenDiE applies sentence-level self-evolution for fine-grained faithfulness improvement (Li et al., 2025). CopyPasteLLM is complementary to these methods: instead of muting internal knowledge or optimizing generic faithful responses, it uses high-copying candidates as a concrete, verifiable preference signal.

Mechanistic views of contextual reliance. Our analysis also connects to mechanistic studies of how transformers use context. Induction heads and related retrieval mechanisms provide a basis for in-context information reuse (Olsson et al., 2022). Recent work argues that instruction finetuning can even hurt context reliance (Goyal et al., 2025). CopyPasteLLM can be interpreted as an alignment strategy that counteracts this inversion by rewarding explicit evidence reuse and discouraging unsupported parametric completion.

6 Analysis

6.1 Why Does Copy-Paste Help?

The central hypothesis is that copying works because it changes the model's default resolution of knowledge conflict. Instead of asking the model to summarize evidence while fighting its prior beliefs, Copy-Paste turns the task into selecting, preserving, and lightly composing spans that are already licensed by the context. This shrinks the space of possible answers and makes unsupported continuation harder. It also gives the evaluator an immediate handle on provenance: copied fragments can be traced back to the source passage without an additional citation model.

6.2 Hidden-State Geometry

Figure 3 compares hidden-state distributions for contextual and parametric trajectories on PubMedQA. For both Mistral-7B-Instruct-v0.2 and Llama-3.1-8B-Instruct, the base model shows more overlap between contextual and parametric representations than CopyPasteLLM. After alignment, contextual traces become more compact, while parametric traces move further away. This is consistent with the intended effect of the method: when the answer should come from the input passage, the model occupies a representational regime that is more distinct from free parametric completion.

6.3 Token-Level Logit Power

Figure 4 reports token-level logit power over response positions. Across RAGTruth, FaithEval, and PubMedQA, CopyPasteLLM assigns stronger positive mass to contextual tokens early in generation, while the magnitude of parametric logits is attenuated. This pattern matches the behavioral results in Tables 2 and 3: the model is not simply copying more surface text, but is changing which knowledge source dominates the next-token distribution.

6.4 Limitations Visible in the Results

The results also show where Copy-Paste is not a universal win. First, the method trades some fluency for faithfulness in Stage 1, especially for stricter variants. Second, it does not uniformly beat every baseline on every metric: Context-DPO remains competitive on seen ConFiQA-QA, and one non-counterfactual PubMedQA result slightly decreases on Llama-3.1-8B. Third, the public repository fully exposes the prompting pipelines but not the full training implementation for Stage 2, so some alignment details in this write-up are reconstructed from the released figures, tables, and model documentation rather than from end-to-end training code. We state this explicitly because the paper's claims should not exceed the public evidence.

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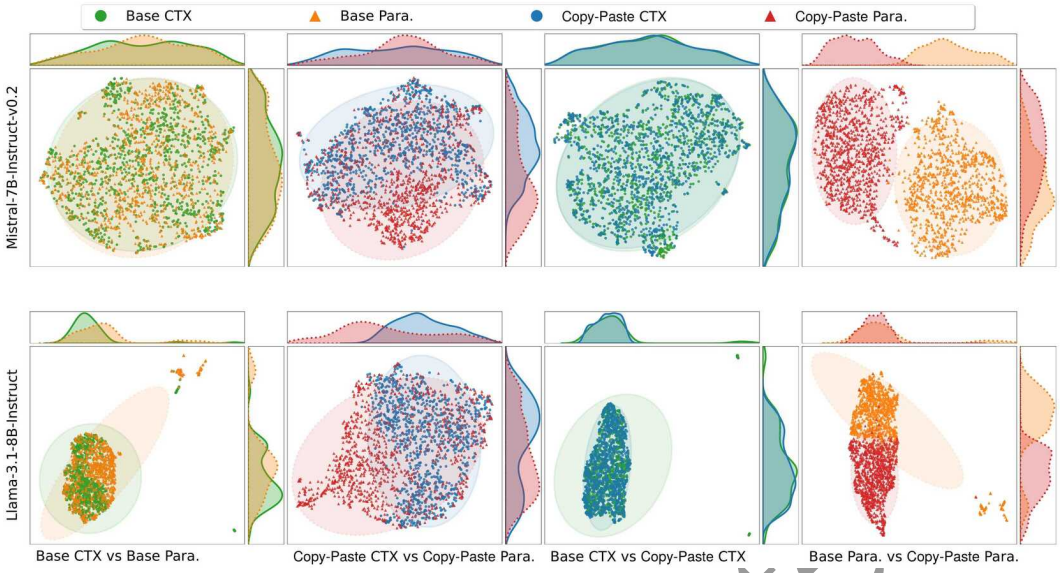

Figure 3: Hidden-state geometry on PubMedQA. CopyPasteLLM yields a clearer separation between contextual and parametric trajectories than the base model on both Mistral-7B-Instruct-v0.2 and Llama-3.1-8B-Instruct, indicating stronger contextual specialization during grounded decoding.

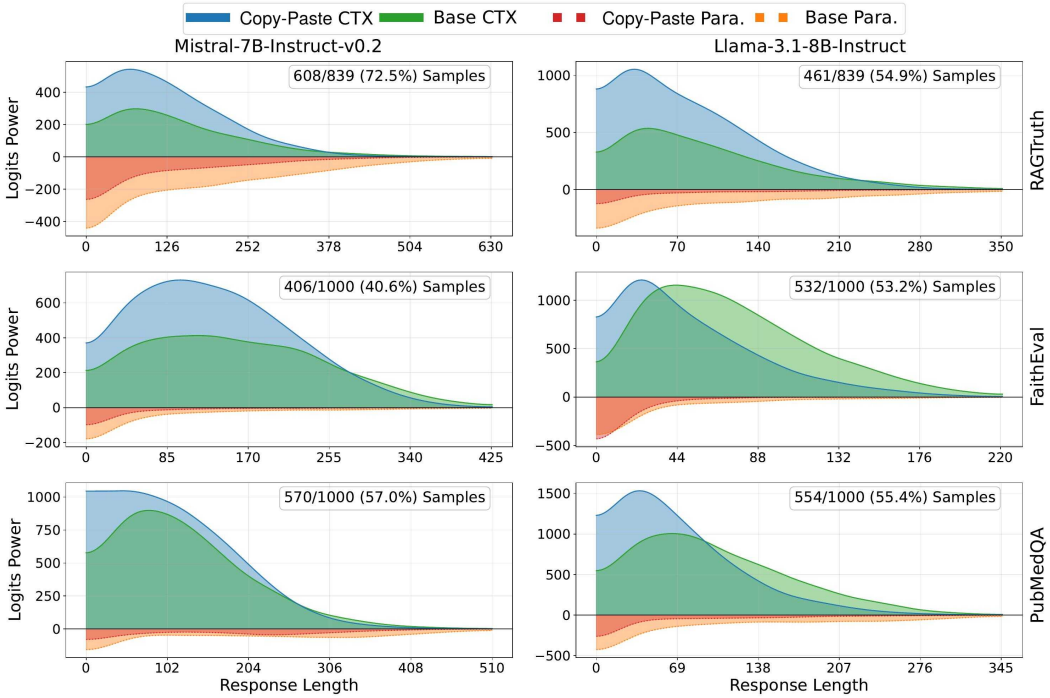

Figure 4: Token-level logit power under contextual and parametric traces. CopyPasteLLM increases contextual contribution early in decoding and reduces the magnitude of parametric competition across datasets and models.

7 Conclusion

CopyPasteLLM starts from a simple empirical observation: responses that reuse more context tend to hallucinate less under knowledge conflict. By turning that observation into both a prompting

strategy and a preference-alignment signal, the method produces large counterfactual faithfulness gains with only 365 training examples, while still improving average accuracy in standard non-counterfactual QA. The mechanistic analyses suggest that these gains arise from a genuine shift toward contextual trust rather than from shallow formatting effects. More broadly, the results support a practical design principle for grounded generation: when faithfulness matters, deliberate copying can be a feature rather than a flaw.

8 Ethics Statement

This work targets a safety-relevant failure mode of LLMs: presenting confident but context-unfaithful answers. The approach is particularly attractive in high-stakes settings such as biomedical QA because copied fragments make evidence provenance easier to inspect. However, Copy-Paste is not a substitute for source verification. A copied answer can still be misleading if the retrieved context itself is wrong, incomplete, or adversarially selected. The method may also over-prefer extractive answers when users actually need synthesis across multiple sources. Deployment should therefore keep source auditing and abstention mechanisms in the loop.

9 Reproducibility Statement

All results reported here are taken from the provided experimental artifacts in the repository: the table files in `tables/`, the figure files in `figures/`, the public implementation of the three prompting pipelines in `code/CopyPasteLLM/`, and the released README describing the model and seed dataset. We preserved the reported numbers exactly, including train sizes and seen-data markers. The implementation publicly exposes the extractiveness metric, the CP-Order/CP-Link/CP-Refine pipelines, and the default refinement settings (maximum five iterations, minimum extractiveness 0.99). The Stage-2 alignment procedure is described from the released pipeline figure and accompanying repository documentation; we therefore report it at the level supported by those artifacts and do not claim additional unpublished implementation details.

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